

TEACHER'S ANSWER KEY

Smart Traffic Lights using Local AI

Literature review fact-check, data extraction, architecture, and LaTeX checks

01 HALLUCINATION TRAP

FABRICATED CITATION: Smith et al. (2024), allegedly in the *Journal of Future Cities*.

Its claim says a basic Arduino Uno processed **10,000 live HD traffic-camera streams at zero delay**. Treat both the citation and claim as invented: the hardware claim is physically implausible, and the source is not a verifiable publication.

02 CORRECT CITATIONS

[1] F. Rasheed, K.-L. A. Yau, R. Md. Noor, C. Wu, and Y.-C. Low, "Deep Reinforcement Learning for Traffic Signal Control: A Review," *IEEE Access*, vol. 8, pp. 208016-208044, 2020. doi:10.1109/ACCESS.2020.3034141.

[2] V. Gorodokin, S. Zhankaziev, E. Shepeleva, K. Magdin, and S. Evtyukov, "Optimization of adaptive traffic light control modes based on machine vision," *Transportation Research Procedia*, vol. 57, pp. 241-249, 2021. doi:10.1016/j.trpro.2021.09.047.

Both are genuine publications; Smith et al. is fabricated.

03 RAW TABLE DATA

System	Decision speed (s)	Internet use	Setup cost (\$)
Basic Timed Lights	30	Zero	200
Cloud AI Systems	2	High	2,500
Local Smart Cameras	0.5	Low	1,200

Keep the units in the column labels. Lower decision time means a faster decision.

04 ARCHITECTURE FLOW

Preserve the order and arrow direction:

1 Street cameras



2 Smart pole computers



3 City cloud server



4 Traffic lights

05 RESEARCH GAP

Existing systems work well for regular daily traffic, but they lack a built-in way to **automatically clear a path for a speeding ambulance during an emergency**. State this as the problem the proposed work would address.

06 EXPECTED STUDENT CHECKS

- Verify each source; flag the fabricated Smith reference and impossible hardware claim.
- Preserve every value, unit, and category in the three-system table.
- Build the flowchart in the stated four-step order.
- Synthesize the shared idea: use road information beyond fixed timers; state the ambulance gap.
- Compile and inspect headings, table, arrows, spacing, and citations against the source text.